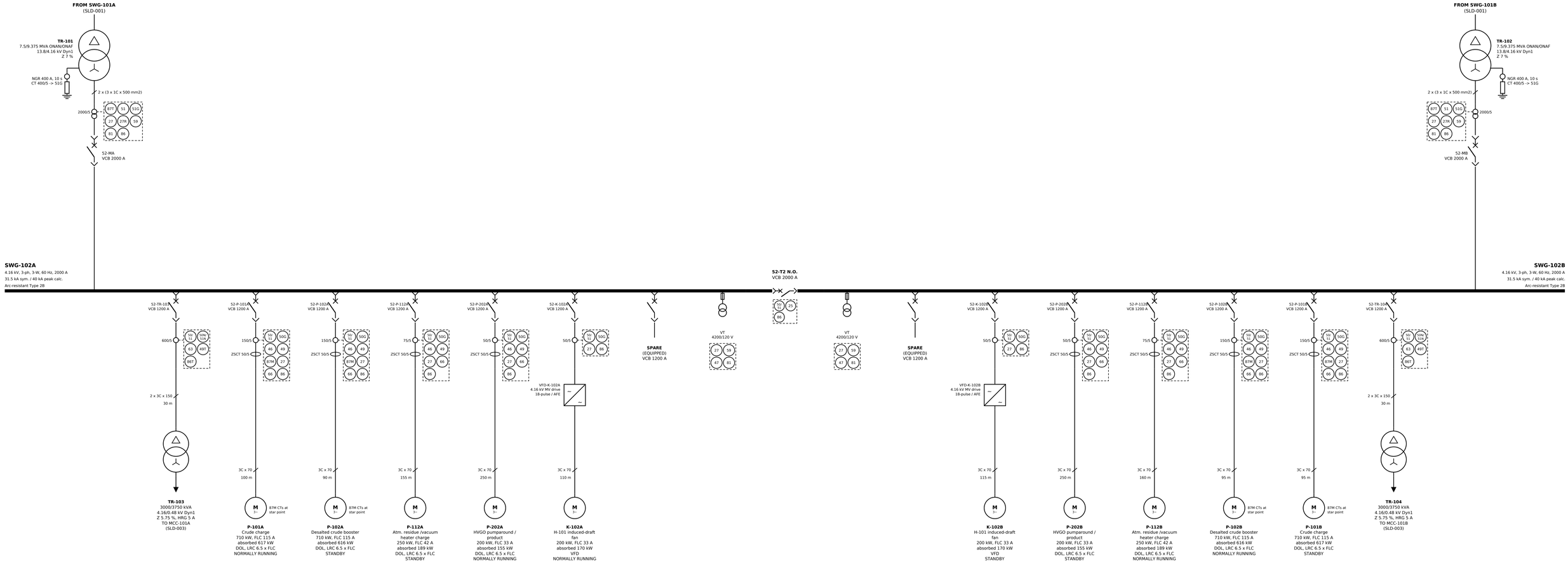


PROTECTIVE DEVICE FUNCTIONS

DEVICE	FUNCTION (ANSI/IEEE C37.2)
25	Synchronism check
27	Undervoltage
27R	Residual voltage (transfer)
46	Negative-sequence / phase unbalance
47	Phase-sequence voltage
49	Thermal overload (motor, RTD)
49T	Transformer winding temperature
50/51	Instantaneous / time overcurrent
50G	Ground fault via ZSCT

DEVICE	FUNCTION
50N/51N	Residual ground overcurrent
51G	Neutral (NGR) ground overcurrent
59	Overvoltage
63	Sudden pressure / Buchholz
66	Starts per hour
67	Directional overcurrent
81	Under/over frequency
86	Lockout relay
87L/M/T	Line / motor / transformer differential

MV MOTOR FEEDER SCHEDULE (term. dlp = bus dlp + cable)

TAG	BUS	kW	FLC A	START	CT	CABLE	L m	VD run %	Vdlp start %
K-102A	SWG-102A	200	33	VFD	50/5	3C x 70 mm2	110	0.05	-
K-102B	SWG-102B	200	33	VFD	50/5	3C x 70 mm2	115	0.06	-
P-101A	SWG-102A	710	115	DOL	150/5	3C x 70 mm2	100	0.17	5.9
P-101B	SWG-102B	710	115	DOL	150/5	3C x 70 mm2	95	0.16	5.8
P-102A	SWG-102A	710	115	DOL	150/5	3C x 70 mm2	90	0.15	5.8
P-102B	SWG-102B	710	115	DOL	150/5	3C x 70 mm2	95	0.16	5.9
P-112A	SWG-102A	250	42	DOL	75/5	3C x 70 mm2	155	0.10	2.3
P-112B	SWG-102B	250	42	DOL	75/5	3C x 70 mm2	160	0.10	2.3
P-202A	SWG-102A	200	33	DOL	50/5	3C x 70 mm2	250	0.12	2.0
P-202B	SWG-102B	200	33	DOL	50/5	3C x 70 mm2	250	0.12	2.0

NOTES

1. Switchgear: 4.16 kV metal-clad, arc-resistant (IEEE C37.20.7 Type 2B), drawout vacuum breakers, 2000 A main bus, 31.5 kA sym. interrupting, 125 V DC control.
2. Motor feeders: VCB + microprocessor motor protection relay (MPR) with RTD module (49). 87M (self-balancing CTs at motor terminal box) for motors >= 500 kW.
3. VFD feeders: MV drive (18-pulse / AFE, IEEE 519 compliant) incl. motor protection (49/46/37).
4. Auto-transfer: on loss of incomer (27 + 27R), incomer trips and tie closes (open transition); manual return. Incomer/tie 2-out-of-3 interlock. MV motors restart by DCS sequence (no auto-restart).
5. CT secondaries 5 A, C200/C400 relaying class; metering 0.3 class. All relays IEC 61850 to ECS.
6. Motor space heaters 120 V fed from MCC-101A/B lighting panels; interlocked with VCB.
7. Cable sizes per CFU-000-EL-CBL-001; lengths are route estimates (layout not final).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

SINGLE LINE DIAGRAM
4.16 KV SWITCHGEAR SWG-102A / SWG-102B

DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	NTS	1 OF 1

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-000-EL-SLD-002**

REV: **A**